

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407343
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lai; Yi-An et al.

Semiconductor devices and circuits with increased breakdown voltage

Abstract

A switching circuit includes a main circuit including a number of first transistors. The main circuit has a first node, a second node, and a third node and is operative in response to a control signal received by the first node, and the second node is configured to receive a supply voltage. The switching circuit also includes an auxiliary circuit electrically coupled to the second node of the main circuit and configured to provide surge protection for the main circuit. The auxiliary circuit includes a second transistor. A breakdown voltage of the second transistor is different than a breakdown voltage of each first transistor of the number of first transistors.

Inventors: Lai; Yi-An (Taipei, TW), Chern; Chan-Hong (Palo Alto, CA), Hsieh; Cheng-Hsiang (Taipei, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsin-Chu, TW)

Family ID: 89018003

Assignee: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.
(Hsinchu, TW)

Appl. No.: 18/629141

Filed: April 08, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240259013 A1	Aug. 01, 2024

Related U.S. Application Data

continuation parent-doc US 17835688 20220608 US 11955956 child-doc US 18629141

Publication Classification

Int. Cl.: H03K17/04 (20060101); H02H9/04 (20060101); H03K17/0812 (20060101)

U.S. Cl.:

CPC H03K17/08128 (20130101); H02H9/04 (20130101); H03K17/08122 (20130101);

Field of Classification Search

CPC: H02H (9/04); H02H (9/00); H02H (9/043); H05K (9/0067); H03K (17/08128); H03K (17/08122); H01L (27/0285)

USPC: 361/56; 361/91.1; 361/111

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8848419	12/2013	Wu et al.	N/A	N/A
9425749	12/2015	Gulko et al.	N/A	N/A
9666683	12/2016	Chiu	N/A	N/A
9780775	12/2016	Brindle et al.	N/A	N/A
2009/0073620	12/2008	Mergens et al.	N/A	N/A
2010/0254051	12/2009	Jeon	361/56	H02H 9/046
2012/0326235	12/2011	Otsuru et al.	N/A	N/A
2014/0078630	12/2013	Koch et al.	N/A	N/A
2015/0092307	12/2014	Petruzzi	N/A	N/A
2019/0199091	12/2018	Handy	N/A	H03K 17/162
2019/0377376	12/2018	Graves	N/A	H01F 29/146
2021/0132137	12/2020	Chen	N/A	N/A
2021/0210481	12/2020	Guan	N/A	N/A
2021/0249404	12/2020	Lai et al.	N/A	N/A
2022/0149037	12/2021	Roig-Guitart	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202131475	12/2020	TW	N/A

Primary Examiner: Nguyen; Danny

Attorney, Agent or Firm: HAYNES AND BOONE, LLP

Background/Summary

PRIORITY DATA (1) The present application is a continuation application of U.S. patent application Ser. No. 17/835,688, filed Jun. 8, 2022, which is herein incorporated by reference in its entirety.

BACKGROUND

- (1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed.
 - (2) In semiconductor technology, Group III-Group V (or III-V) semiconductor compounds (e.g., gallium nitride (GaN)) may be used to form various integrated circuit (IC) devices, such as high-power field-effect transistors (FETs), high frequency transistors, or high-electron-mobility transistors (HEMTs). A high-electron-mobility transistor (HEMT) is a field effect transistor having a 2-dimensional electron gas (2DEG) layer close to a junction between two materials with different band gaps (i.e., a heterojunction). The 2-DEG layer is used as the transistor channel instead of a doped region, as is generally the case for metal oxide semiconductor field effect transistors (MOSFETs). Compared with MOSFETs, HEMTs have a number of attractive properties such as high breakdown voltage and low on-resistance. In some examples, due to its high breakdown voltage and low on-resistance, GaN-based HEMT may be used in an integrated circuit (e.g., switching power supplies). However, turning off the switches in the switching power supplies may create voltage spikes, also known as surges. To prevent the damages caused by the voltage spikes, there is a need to further increase the breakdown voltage of the GaN-based HEMT. Accordingly, improvements in this area are needed.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 illustrates a schematic of an exemplary simplified switching circuit having a main circuit and an auxiliary circuit, according to various aspects of the present disclosure.
- (3) FIG. 2 illustrates a schematic of an exemplary simplified main circuit of the switching circuit shown in FIG. 1, according to various aspects of the present disclosure.
- (4) FIG. 3 illustrates an exemplary cross-sectional view of a first transistor implemented in the main circuit shown in FIG. 2, according to various aspects of the present disclosure.
- (5) FIG. 4 illustrates an exemplary cross-sectional view of a second transistor implemented in the auxiliary circuit shown in FIG. 1, according to various aspects of the present disclosure.
- (6) FIG. 5 illustrates a fragmentary top view of the first transistor, according to various aspects of the present disclosure.
- (7) FIG. 6 illustrates a fragmentary top view of the second transistor, according to various aspects of the present disclosure.
- (8) FIG. 7 illustrates simulated timing diagrams showing voltages over time or current over time at different nodes of the switching circuit shown in FIG. 1.
- (9) FIG. 8 illustrates a flow chart of a method for configuring a switch circuit to have an increased breakdown voltage.

(10) FIG. 9 illustrates a schematic of another exemplary simplified switching circuit having the main circuit and a different auxiliary circuit, according to various aspects of the present disclosure. (11) FIG. 10 illustrates simulated timing diagrams showing the voltages over time or current over time at different nodes of the switching circuit shown in FIG. 9.

DETAILED DESCRIPTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(13) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(14) Further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range considering variations that inherently arise during manufacturing as understood by one of ordinary skill in the art. For example, the number or range of numbers encompasses a reasonable range including the number described, such as within $\pm 10\%$ of the number described, based on known manufacturing tolerances associated with manufacturing a feature having a characteristic associated with the number. For example, a material layer having a thickness of “about 5 nm” can encompass a dimension range from 4.25 nm to 5.75 nm where manufacturing tolerances associated with depositing the material layer are known to be $\pm 15\%$ by one of ordinary skill in the art. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(15) Compared with MOSFETs, HEMTs have a number of attractive properties such as high breakdown voltage and low on-resistance, and thus, HEMTs are widely used in various applications. In some embodiments, a switching circuit that includes a number of switch transistors (e.g., HEMTs) may be used in a power conversion circuit (e.g., a DC-DC converter). For example, a number of switch transistors may include HEMTs and may be coupled to a power supply. Turning off those switch transistors may lead to voltage spikes or surges, thereby damaging those switch transistors. For example, each switch transistor may have a breakdown voltage that is about, for example, 650V, and the surge voltage may be about, for example, 800V. Breakdown voltage of those HEMTs may be increased to prevent those HEMT-based switch transistors from being damaged by the voltage spikes. However, increasing the breakdown voltage of those HEMTs from 650V to 800V may include introducing a complicated field plate design and/or increasing a drift region length (i.e., a distance between gate structure and drain feature of the HEMTs), which may provide an increased fabrication cost associated with the formation of the HEMTs or an increased footprint for each HEMT of those HEMTs, thereby taking up an undue amount of real estate in an IC chip.

(16) The present embodiments are directed to methods and circuits that provide surge protection to those HEMTs without increasing the breakdown voltage of the HEMTs. In an embodiment, a switching circuit includes a number of HEMTs connected in parallel and coupled to a power supply. Each of those HEMTs has a first breakdown voltage **BV1**. The power supply may generate a surge voltage that is greater than the first breakdown voltage **BV1**. An auxiliary circuit is electrically coupled to the switching circuit to provide surge protection. In an embodiment, the auxiliary circuit includes a HEMT having a second breakdown voltage **BV2** greater than the first breakdown voltage **BV1**. When a surge voltage is generated, the surge voltage may be discharged by the HEMT in the auxiliary circuit. By providing the auxiliary circuit, without increasing the first breakdown voltage **BV1**, HEMTs in the switching circuit are protected from damage due to voltage overshoot spikes. The various aspects of the present disclosure will now be described in more detail with reference to the figures.

(17) FIG. 1 illustrates a schematic of an exemplary simplified switching circuit **100** having an auxiliary circuit **130**, according to various aspects of the present disclosure. In embodiments represented in FIG. 1, the switching circuit **100** includes a main circuit **110**. The main circuit **110** has a first node **110a** electrically coupled to a drive circuit **120** and configured to receive a control signal from the drive circuit **120**. The main circuit **110** also has a second node **110b** configured to receive a supply voltage V_{pp} from a power line **115** (or a power supply **115**). The main circuit **110** also includes a third node **110c** configured to receive a reference voltage (e.g., ground voltage **GND**). In the present embodiment, the third node **110c** is coupled to a ground voltage. In some embodiments, the main circuit **110** includes a number of switch transistors (e.g., switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** shown in FIG. 2). Those switch transistors may have a same breakdown voltage **BV1**. An exemplary schematic of the main circuit **110** is described in further detail with reference to FIG. 2. As described above, turning off those switch transistors in the main circuit **110** may introduce a surge voltage to the power line **115**. For example, the power line **115** is configured to provide a normal level of voltage **V1**, and in situations where those switch transistors are turned off, a surge voltage **V2** that is greater than the voltage **V1** and greater than the breakdown voltage **BV1** may be created. That is, the supply voltage V_{pp} provided by the power line **115** may be equal to the normal level of voltage **V1** or the surge voltage **V2**. If the surge voltage **V2** is fully applied to the second node **110b** of the main circuit **110**, since the surge voltage **V2** is greater than the breakdown voltage **BV1**, those switch transistors in the main circuit **110** may be damaged. To prevent the main circuit **110** from being damaged by the surge voltage **V2**, an auxiliary circuit **130** is electrically coupled to main circuit **110** to provide a surge protection.

(18) The auxiliary circuit **130** includes a capacitor **140**. One terminal **140a** of the capacitor **140** is electrically coupled to the second node **110b** of the main circuit **110**. In an embodiment, a capacitance of the capacitor **140** may be between about 1 pF and about 100 nF. The auxiliary circuit **130** also includes a resistor **150** electrically coupled to the other terminal **140b** of the capacitor **140a**. In the present embodiments, the other terminal of the resistor **150** is configured to receive a reference voltage (e.g., ground voltage **GND**). In an embodiment, a resistance of the resistor **150** may be between about 1 K Ω and about 100 K Ω .

(19) The auxiliary circuit **130** also includes a transistor **160**. The transistor **160** has a first terminal **160a** (e.g., gate terminal) electrically coupled to the terminal **140b** of the capacitor **140**, a second terminal **160b** electrically coupled to the second node **110b** of the main circuit **110**, and a third terminal **160c** configured to receive a reference voltage (e.g., ground voltage **GND**). In an embodiment, the transistor **160** includes a GaN-based HEMT and has a breakdown voltage **BV2** that is higher than the breakdown voltage **BV1** of the switching transistors in the main circuit **110**. It is understood that transistor **160** is not limited to a GaN-based HEMT. In an embodiment, an operation voltage of the transistor **160** is higher than an operation voltage of each of the switch transistors in the main circuit **110**. When the switching transistors in the main circuit **110** are turned off and a surge voltage **V2** is generated, or there is a spike voltage **V2** from system, a surge current

will flow through the capacitor **140** and build up a voltage by the resistor **150**. This built-up voltage may then turn on the transistor **160**. Thus, the surge voltage/spike voltage **V2** may be discharged by the transistor **160**. An exemplary cross-sectional view of a structure of the transistor **160** will be described in detail with reference to FIG. 4. In some embodiments, by providing different transistors **160** (having different breakdown voltages) for the auxiliary circuit **130**, the switching circuit **100** may be configured to sustain different surge voltages, and thus the switching circuit **100** may be implemented in different applications.

(20) FIG. 2 illustrates a schematic of an exemplary simplified main circuit **110** of the switching circuit **100** shown in FIG. 1, according to various aspects of the present disclosure. In embodiments represented in FIG. 2, the main circuit **110** includes a number of switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** in parallel connection. **N** is an integer and may be greater than 1000. The switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** are connected in parallel. More specifically, each of the switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** includes a gate terminal, a drain terminal, and a source terminal. The gate terminals of those switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** are electrically coupled to an output of the drive circuit **120** and are configured to receive the control signal from the drive circuit **120**, the drain terminals of those switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** are electrically coupled to the power line **115**, and the source terminals of those switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** are configured to receive, for example, a ground voltage. In an embodiment, each of the switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** includes a GaN-based HEMT, and each HEMT has the same structure and configuration, and thus has the same breakdown voltage **BV1**. For example, those HEMTs in the main circuit **110** have the same gate width **Wg1** (shown in FIG. 5) and same distance between the gate structure and its respective drain feature (i.e., **Lgd1** shown in FIG. 3 and FIG. 5). The distance between the gate structure and its respective drain feature of the HEMT based transistor **160** in the auxiliary circuit **130** may be referred to as **Lgd2** (shown in FIG. 4 and FIG. 6) and is greater than the distance **Lgd1**. A fragmentary cross-sectional view of an exemplary structure of the switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** will be described in further detail with reference to FIG. 3. It is understood that each of the switch transistors **110.sub.1**, **110.sub.2**, . . . **110.sub.N** is not limited to a GaN-based HEMT.

(21) In an embodiment, the first breakdown voltage **BV1** may be about 500V, and the second breakdown voltage **BV2** may be about 800V, and the main circuit **110** may include 3000 switch transistors, a total gate width of the 3000 switch transistors may be about 300 mm. To prevent the main circuit **110** from being damaged due to surge voltage, instead of providing the auxiliary circuit **130**, another possible method may include increasing the length of the drain drift region (i.e., the region between gate structure and its respective drain feature) of each switch transistor of those 3000 switch transistors in the main circuit **110** from **Lgd1** to **Lgd2**. However, increasing the length of the drain drift region of each switch transistor from **Lgd1** to **Lgd2** may significantly and disadvantageously increase a total footprint of the main circuit **110**. For example, a total chip area of a main circuit that includes switch transistors each having the second breakdown voltage **BV2** may be about 7 mm.^{sup.2}; however, a total chip area of a main circuit that includes switch transistors each having the first breakdown voltage **BV1** may be about 4.2 mm.^{sup.2} and a total chip area of the auxiliary circuit may be about 0.02 mm.^{sup.2}. Therefore, compared with embodiments where the length of the drain drift region of each switch transistor is increased from **Lgd1** to **Lgd2** to increase the breakdown voltage from **BV1** to **BV2**, the switching circuit **100** that implements the auxiliary circuit **130** may have a smaller footprint (reduced by 40%) and takes less amount of real estate of an IC chip.

(22) FIG. 3 illustrates an exemplary cross-sectional view of the switch transistor **110.sub.1** implemented in the main circuit **110** shown in FIG. 2, according to various aspects of the present disclosure. Since the switch transistor **110.sub.1** includes a HEMT, the switch transistor **110.sub.1** may be referred to as a semiconductor device **110.sub.1** or a HEMT **110.sub.1**. The semiconductor

device **110.sub.1** includes a substrate **310**. The substrate **310** may include silicon carbide (SiC), sapphire, or silicon (Si). In the present embodiment, the substrate **310** is a silicon substrate.

(23) The semiconductor device **110.sub.1** also includes a nucleation layer **320** formed over the substrate **310**. The nucleation layer **320** has a lattice structure and/or a thermal expansion coefficient (TEC) suitable for, for example, bridging the lattice mismatch and/or the TEC mismatch between the substrate **310** and a layer thereover. In some embodiments, the nucleation layer **320** includes aluminum nitride (AlN).

(24) The semiconductor device **110.sub.1** also includes a buffer layer **330** formed over the nucleation layer **320**. In some embodiments, the buffer layer **330** includes a graded aluminum-gallium nitride (Al.sub.xGa.sub.1-xN, x is the aluminum content ratio in the aluminum-gallium constituent, 0<x<1) layer. In some embodiments, the buffer layer **330** may include multiple aluminum gallium nitride layers with different x ratios. In some other embodiments, instead of having multiple layers with different x ratios, the buffer layer **330** may have a continuous gradient of the ratio x.

(25) The semiconductor device **110.sub.1** also includes a super lattice structure **340** formed over the buffer layer **330**. The super lattice structure **340** may include a number of first group III-V layers and a number of second group III-V layers (not separately labeled) vertically and alternately stacked, and the first group III-V layers have a lattice constant different than the second group III-V layers. For example, the first group III-V layers may include AlN, and the second group III-V layers may include GaN.

(26) The semiconductor device **110.sub.1** also includes a channel layer **350** formed over the super lattice structure **340**. In some embodiments, the channel layer **350** may include one or more Group III-V compound layers such as GaN, AlGa_N, InGa_N and InAlGa_N. In one embodiment, the channel layer **350** includes a GaN layer.

(27) The semiconductor device **110.sub.1** also includes an active layer **360** formed over the channel layer **350**. The active layer **360** includes one or more Group III-V compound layers which are different from the Group III-V compound layers of the channel layer **350** in composition. In an embodiment, the active layer **360** may include AlGa_N. The active layer **360** is configured to cause a 2-dimensional electron gas (2DEG) to be formed in the channel layer **350** along an interface between the channel layer **350** and the active layer **360**. A heterojunction is formed between the active layer **360** and the channel layer **350** having two different semiconductor materials.

(28) The semiconductor device **110.sub.1** also includes a source feature **365** and a drain feature **370** disposed over the active layer **360**. In the present embodiments, the source feature **365** and drain feature **370** are formed to be in ohmic contact with an upper surface of the active layer **360**. In some situations, the source feature **365** may be referred to as a source contact **365**, and the drain feature **370** may be referred to as a drain contact **370**. In some embodiments, the source contact **365** and the drain contact **370** may have the same composition and may include a metal layer that include titanium (Ti), titanium nitride (TiN), aluminum copper (AlCu) alloy, combinations thereof, or other suitable materials.

(29) The semiconductor device **110.sub.1** also includes a gate structure **375** disposed over the active layer **360** and between the source feature **365** and the drain feature **370**. The gate structure **375** includes a gate dielectric layer and a gate electrode layer disposed over the gate dielectric layer. The gate electrode includes a conductive material layer. In various examples, the conductive material layer may include nickel (Ni), gold (Au) copper (Cu), titanium (Ti), titanium nitride (TiN), titanium tungsten (TiW), titanium tungsten nitride (TiWN), tungsten (W) or tungsten nitride (WN), combinations thereof, or other suitable materials. In the present embodiments, along the X direction, a distance Lgd1 between the gate structure **375** and the drain contact **370** is greater than a distance between the gate structure and the source contact **365**. A dielectric layer **380** is formed over the source feature **365**, the drain feature **370**, and the gate structure **375**. The dielectric layer **380** may include multiple layers and each layer may include silicon oxide (SiO.sub.2), silicon

nitride (SiN), combinations thereof, or other suitable materials. The semiconductor device **110.sub.1** also includes contact vias penetrating through the dielectric layer **380** and in direct contact with the source feature **365**, the drain feature **370**, or the gate structure **375**. For example, the semiconductor device **110.sub.1** includes a contact via **385** extending through the dielectric layer **380** and in direct contact with the source feature **365**. In some embodiments, the semiconductor device **110.sub.1** may also include one or more field plates (e.g., field plate **390**) disposed over the dielectric layer **380**. In the present embodiments, the semiconductor device **110.sub.1** includes a field plate **390** disposed over the dielectric layer **380** and electrically coupled to the source feature **365** via the contact via **385**.

(30) FIG. 4 illustrates an exemplary cross-sectional view of the transistor **160** implemented in the auxiliary circuit **1630** shown in FIG. 1, according to various aspects of the present disclosure. In the present embodiments, a structure of the transistor **160** is in a way similar to that of the switch transistor **110.sub.1**. For example, the transistor **160** also includes the substrate **310**, the nucleation layer **320** formed over the substrate **310**, the buffer layer **330** formed over the nucleation layer **320**, the super lattice structure **340** formed over the buffer layer **330**, the channel layer **350** formed over the super lattice structure **340**, and the active layer **360** formed over the channel layer. The transistor **160** also includes a source feature **465** and a drain feature **470** disposed over the active layer **360**, and a gate structure **475** disposed between the source feature **465** and the drain feature **470**. A distance L_{gd2} between the gate structure **475** and the drain feature **470** is greater than a distance between the gate structure **475** and the source feature **465**. Since the auxiliary circuit **130** is configured to provide surge protection for the transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110**, a breakdown voltage BV_2 of the transistor **160** is higher than a breakdown voltage BV_1 of the switch transistor **110.sub.1**. In the present embodiments, the distance L_{gd2} between the gate structure **475** and the drain feature **470** is greater than the distance L_{gd1} between the gate structure **375** and the drain feature **370** such that the transistor **160** has a higher breakdown voltage than the switch transistor **110.sub.1**.

(31) FIG. 5 illustrates a fragmentary top view of the switch transistor **110.sub.1**, according to various aspects of the present disclosure. In the present embodiments, the gate structure **375** extends along the Y direction and has a gate width W_{g1} along the Y direction. In an embodiment, the gate width W_{g1} is between about 50 μm and about 150 μm . The distance L_{gd1} may be configured accordingly with respect to a satisfactory breakdown voltage BV_1 of the switch transistor **110.sub.1**. In an embodiment, the breakdown voltage BV_1 of the switch transistor **110.sub.1** may be about 500V, and the distance L_{gd1} may be between about 6 μm and about 18 μm . The gate width W_{g1} and the distance L_{gd1} may affect a footprint of the transistor **110.sub.1**.

(32) FIG. 6 illustrates a fragmentary top view of the transistor **160**, according to various aspects of the present disclosure. In the present embodiments, the gate structure **475** extends along the Y direction and has a gate width W_{g2} along the Y direction. In an embodiment, W_{g2} is between about 50 μm and about 150 μm . The distance L_{gd2} is greater than the distance L_{gd1} . The distance L_{gd2} may be configured with respect to a satisfactory breakdown voltage BV_2 of the transistor **160** and is no more than 1000 μm . If the distance L_{gd2} is greater than 1000 μm , a larger current is needed to turn on the transistor **160**, more time may be needed to charge the capacitor to generate this larger current, and the transistor **160** may thus not be able to discharge the surge voltage in time.

(33) In an embodiment, the breakdown voltage BV_2 of the transistor **160** may be about 800V, and the distance L_{gd2} is greater than the distance L_{gd1} and may be between about 9 μm and about 27 μm . In an embodiment, to provide the transistor **160** a higher breakdown voltage (e.g., 1000V), the distance L_{gd2} may be increased and between about 12 μm and about 35 μm . In an embodiment, the distance L_{gd2} may be greater than 23 μm . The gate width W_{g2} and the distance L_{gd2} may affect a footprint of the transistor **160**. In some embodiments, the gate width W_{g2} may be equal to the gate width W_{g1} . In an embodiment, a footprint of the transistor **160** is greater than a footprint of each

switch transistor of the switch transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110**, and the footprint of the transistor **160** is less than a total footprint of the switch transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110**.

(34) FIG. 7 illustrates simulated timing diagrams **710**, **720**, **730**, and **740** showing voltages over time or current over time at different nodes of the switching circuit **100** shown in FIG. 1. More specifically, timing diagram **710** represents a voltage signal V_{pp} provided by the power line **115**, timing diagram **720** represents a voltage signal V_{dd} measured at the second node **110b** of the main circuit **110**, timing diagram **730** represents a voltage signal V_g measured at the first node **110a** of the main circuit **110**, and timing diagram **740** represents a current I_{rq} that flows through the transistor **160**.

(35) In this present simulation, the power line **115** is configured to provide a voltage V_{pp} . A normal level V_1 of the voltage V_{pp} is ranged between about 300V and about 500V, the switch transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110** each has the distance L_{gd1} (shown in FIG. 3 and FIG. 5), each transistor in the main circuit **110** has an operation voltage that is about 400V and a breakdown voltage that is about 500V, and the transistor **160** in the auxiliary circuit **130** has a distance L_{gd2} (shown in FIG. 4 and FIG. 6), operation voltage that is about 650V, and a breakdown voltage that is about 800V. When the switch transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110** are turned off, as represented by the timing diagram **730**, the power line **115** provides surge voltages V_2 . The surge voltages may be greater than 800V. By providing the auxiliary circuit **130**, as represented by the timing diagram **720**, the voltage V_{dd} that is supplied to the main circuit **110** may be between about 395V and about 405V, which is less than the breakdown voltage of the transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110**, and the surge voltage is discharged by the transistor **160**, as presented by the timing diagram **740**. Therefore, providing the auxiliary circuit **130** to increase the breakdown voltage of the switching circuit **100** may advantageously prevent the switch transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110** from being damaged by the surge voltage V_2 . Besides providing surge protection, compared with embodiments where the breakdown voltage of the transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110** are increased to prevent damages due to surge voltage, the implementation of the auxiliary circuit **130** also advantageously reduces a total footprint of the switching circuit **100**.

(36) FIG. 8 illustrates a flow chart of a method **800** for configuring a switch circuit to have an increased breakdown voltage. In an embodiment, the method **800** includes, at **802**, determining a configuration of switch transistors in a main circuit. For example, a breakdown voltage BV_1 and the length L_{gd1} of each switch transistor of the transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110** may be determined. The method **800** also includes, at **804**, estimating a surge voltage/spike voltage (e.g., V_2) of a power line (e.g., power line **115**) that is configured to provide power supply to the main circuit **110**. The method **800** also includes, at **806**, in response to the configuration of the main circuit and the surge/spike voltage, determining a configuration of an auxiliary circuit to provide a surge/spike protection for the main circuit **110**. For example, based on the surge voltage V_2 and the breakdown voltage BV_1 of the switch transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110**, a breakdown voltage BV_2 of the transistor **160** in the auxiliary circuit **130** may be determined. The method **800** also includes, at **808**, in response to the determined breakdown voltage BV_2 of the transistor **160**, determining a configuration of the transistor **160**. For example, a distance L_{gd2} between a gate structure and a drain feature of the transistor **160** may be determined such that the transistor **160** has the breakdown voltage BV_2 . As such, the auxiliary circuit **130** may be configured to provide surge/spike protection for the main circuit **110**. In other words, the breakdown voltage of the switching circuit **100** is increased by the auxiliary circuit **130**.

(37) FIG. 9 illustrates a schematic of another exemplary simplified switching circuit **100'** having an auxiliary circuit **130'**, according to various aspects of the present disclosure. In the present

embodiments, the switching circuit **100'** includes the main circuit **110** having the first node **110a** electrically coupled to the drive circuit **120**, the second node **110b** electrically coupled to the power line **115**, and the third node **110c** electrically coupled to a reference voltage such as a ground voltage. As described above with reference to FIGS. **1** and **2**, the main circuit **110** includes a number of switch transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** connected in parallel.

(38) The switching circuit **100'** also includes an auxiliary circuit **130'** electrically coupled to the second node **110b** of the main circuit **110**. The auxiliary circuit **130'** includes the capacitor **140**. One terminal **140a** of the capacitor **140** is electrically coupled to the second node **110b** of the main circuit **110**. The auxiliary circuit **130'** also includes the resistor **150** electrically coupled to the other terminal **140b** of the capacitor **140a**. The auxiliary circuit **130'** also includes the transistor **160**. The first terminal **160a** (e.g., gate terminal) of the transistor **160** is electrically coupled to the terminal **140b** of the capacitor **140**, the second terminal **160b** (e.g., drain terminal) of the transistor **160** is electrically coupled to the second node **110b** of the main circuit **110**, and the third terminal **160c** is configured to receive a reference voltage (e.g., ground voltage GND). In an embodiment, the transistor **160** includes a GaN-based HEMT and has a breakdown voltage BV2 that is higher than the breakdown voltage BV1 of the switching transistors in the main circuit **110**. Thus, when the switching transistors in the main circuit **110** are turned off and when a surge voltage V2 is generated, the capacitor **140** may turn on the transistor **160** and thus the surge voltage V2 may be discharged by the transistor **160**.

(39) As described above, when switch transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110** are turned off, a surge voltage V2 that is greater than the voltage V1 and greater than the breakdown voltage BV1 may be created. The gate terminal (i.e., the first terminal **160a**) of the transistor **160** is susceptible to damage due to voltage overshoot spikes that exceed its gate breakdown voltage. The auxiliary circuit **130'** includes a gate protection circuit **170** electrically coupled to the terminal **140b** of the capacitor **140** and the first terminal **160a** of the transistor **160**. More specifically, the gate protection circuit **170** includes one terminal connected to the terminal **140b** of the capacitor **140**, and the other terminal configured to receive a reference voltage (e.g., a ground voltage). In the present embodiments, the gate protection circuit **170** includes a number of diodes **170.sub.1**, . . . , **170.sub.M**, connected in serial. M is an integer and is greater than 1. In some embodiments, M may be between 2 and 10, depending on the threshold voltage of each of the diodes **170.sub.1**, . . . , **170.sub.M** and the gate breakdown voltage of the transistor **160**. Current that is generated due to the surge event may partially flow through the path **170p**. By implementing the gate protection circuit **170**, a gate input voltage of the transistor **160** may be clamped during a surge event, protecting the gate terminal of the transistor **160** from being damaged. In some embodiments, the gate protection circuit **170** may consume portions of the surge voltage not consumed by the resistor **150** or the transistor **160**.

(40) FIG. **10** illustrates simulated timing diagrams **1010**, **1020**, **1030**, **1040**, and **1050** showing the voltage or current over time at different nodes of the switching circuit shown in FIG. **9**. More specifically, timing diagram **1010** represents a voltage signal Vpp provided by the power line **115**, timing diagram **1020** represents a voltage signal Vdd measured at the second node **110b** of the main circuit **110**, timing diagram **930** represents a voltage signal Vg measured at the first node **110a** of the main circuit **110**, timing diagram **1040** represents a voltage signal Vr1 measured at the terminal **140b** of the capacitor **140**, and timing diagram **1050** represents a current Irq that flows through the transistor **160**. The timing diagrams **1010**, **1020**, **1030**, and **1050** are in a way similar to the timing diagrams **710**, **720**, **730**, and **740** described with reference to FIG. **7** and repeated description is omitted for reason of simplicity. When the transistors **110.sub.1**, **110.sub.2**, . . . , **110.sub.N** in the main circuit **110** are turned on or off, as represented by the timing diagram **1040**, current that is generated due to the surge event may flow through the path **170p**, protecting the gate terminal of the transistor **160** from being damaged.

(41) Although not intended to be limiting, one or more embodiments of the present disclosure

provide many benefits to a switching circuit. In an embodiment, a switching circuit includes a main circuit and an auxiliary circuit electrically coupled to the main circuit. The auxiliary circuit includes a power device (e.g., GaN-based HEMT) and is configured to provide surge protection for the main circuit. The main circuit may include a number of transistors that have a breakdown voltage less than a breakdown voltage of the power device. Therefore, without significantly consuming an amount of real estate of an IC chip, the main circuit may be protected from damage due to voltage overshoot spikes. In some embodiments, the breakdown voltage of the power device may be adjusted by changing a distance between a gate structure and a drain feature of the power device. As such, by providing the power device different breakdown voltages, without changing the configuration of the main circuit, the switching circuit may be designed to be implemented for various applications having different surge voltages. In some embodiments, a gate protection circuit is electrically coupled to the power device to protect the gate terminal of the power device. In some embodiments, the auxiliary circuit may be implemented to provide surge protection for other circuits that can adjust the distance between the gate structure and the drain feature to design breakdown voltages. The auxiliary circuit may be readily integrated into existing HEMTs and circuits.

(42) The present disclosure provides for many different embodiments. Semiconductor structures and methods of fabrication thereof are disclosed herein. In one exemplary aspect, the present disclosure is directed to a switching circuit. The switching circuit includes a main circuit including a plurality of first transistors and having a first node, a second node, and a third node, where the main circuit is operative in response to a control signal received by the first node, and the second node is configured to receive a supply voltage. The switching circuit also includes an auxiliary circuit electrically coupled to the second node of the main circuit and configured to provide surge protection for the main circuit, where the auxiliary circuit comprises a second transistor. A breakdown voltage of the second transistor is different than a breakdown voltage of each first transistor of the plurality of first transistors.

(43) In some embodiments, the breakdown voltage of the second transistor may be greater than the breakdown voltage of each first transistor of the plurality of first transistors. In an embodiment, each first transistor of the plurality of first transistors may include a gate terminal electrically coupled to the first node, a drain terminal electrically coupled to the second node, and a source terminal electrically coupled to the third node. In an embodiment, the auxiliary circuit may also include a capacitive element comprising a first terminal and a second terminal, where the first terminal of the capacitive element may be electrically coupled to the second node. The auxiliary circuit may also include a resistive element comprising a third terminal and a fourth terminal, the third terminal being electrically coupled to the second terminal, and the fourth terminal being electrically coupled to a ground voltage. In an embodiment, a gate terminal of the second transistor may be electrically coupled to the second terminal of the capacitive element, a drain terminal of the second transistor may be electrically coupled to the second node, and a source terminal of the second transistor may be electrically coupled to a ground voltage. In an embodiment, the switching circuit may also include a gate protection circuit electrically coupled to the second terminal of the capacitive element, where the gate protection circuit may include a plurality of diodes connected in serial. In an embodiment, the first transistor may be a first III-V based high-electron-mobility transistor (HEMT) and the second transistor may be a second III-V based HEMT. In an embodiment, a substrate of the first III-V based HEMT and a substrate of the second III-V based HEMT may include silicon. In an embodiment, the first III-V based HEMT may be a first gate structure and a first drain feature, and the first gate structure may be spaced apart from the first drain feature by a first distance along a first direction, and the second III-V based HEMT may be a second gate structure and a second drain feature, and the second gate structure may be spaced apart from the second drain feature by a second distance along the first direction, where the second distance may be greater than the first distance. In an embodiment, a width of the first gate structure

along a second direction may be equal to a width of the second gate structure along the second direction, the second direction being substantially perpendicular to the first direction.

(44) In another exemplary aspect, the present disclosure is directed to a circuit. The circuit includes a drain voltage input terminal configured to receive a first voltage, a source voltage input terminal configured to receive a second voltage, a main circuit connected between the drain voltage input terminal and the source voltage input terminal and comprising a plurality of first transistors in parallel connection, a second transistor comprising a drain terminal electrically coupled to the drain voltage input terminal, a source terminal electrically coupled to the source voltage input terminal, and a gate terminal, a capacitive element comprising a first terminal and a second terminal, the first terminal of the capacitive element being electrically coupled to the drain voltage input terminal, and a resistive element comprising a third terminal and a fourth terminal, the third terminal being electrically coupled to the second terminal, and the fourth terminal being electrically coupled to the source voltage input terminal, where the gate terminal of the second transistor is electrically coupled to the second terminal of the capacitive element.

(45) In some embodiments, a breakdown voltage of the second transistor may be greater than a breakdown voltage of each first transistor of the plurality of first transistors. In some embodiments, a sum of the breakdown voltage of the second transistor and the breakdown voltage of each first transistor of the plurality of first transistors may be greater than a surge voltage associated with the first voltage. In some embodiments, an operation voltage of the second transistor may be greater than an operation voltage of each first transistor of the plurality of first transistors. In some embodiments, a footprint of the second transistor may be greater than a footprint of each first transistor of the plurality of first transistors and may be smaller than a total footprint of the plurality of first transistors. In some embodiments, the circuit may also include a plurality of diodes connected in serial and connected between the second terminal of the capacitive element and the source voltage input terminal.

(46) In yet another exemplary aspect, the present disclosure is directed to a circuit. The circuit includes a first circuit configured to receive a power supply voltage from a power line and comprising a plurality of first power devices connected in parallel, wherein each first power device of the plurality of first power devices comprises a first breakdown voltage, a second circuit configured to provide surge protection to the first circuit and comprising a second power device having a second breakdown voltage, where the first breakdown voltage is less than the second breakdown voltage.

(47) In some embodiments, the second circuit may also include a capacitive element comprising a first terminal and a second terminal, the first terminal of the capacitive element being electrically coupled to the power line, and a resistive element comprising a third terminal and a fourth terminal, the third terminal being electrically coupled to the second terminal, and the fourth terminal being electrically coupled to a ground voltage, where a gate terminal of the second power device may be electrically coupled to the second terminal of the capacitive element, a drain terminal of the second power device may be electrically coupled to the power line, and a source terminal of the second power device may be electrically coupled to the ground voltage. In some embodiments, the circuit may also include a gate protection circuit electrically connected between the second terminal of the capacitive element and the ground voltage, where the gate protection circuit may include a plurality of diodes connected in serial. In some embodiments, each of the plurality of first power devices may include a gate structure spaced from a drain feature by a first distance, the second power device may include a gate structure spaced from a drain feature by a second distance, where the second distance may include greater than the first distance.

(48) The foregoing outlines features of several embodiments so that those of ordinary skill in the art may better understand the aspects of the present disclosure. Those of ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the

same advantages of the embodiments introduced herein. Those of ordinary skill in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A switching circuit comprising: a main circuit comprising a plurality of first transistors and comprising a first node, a second node, and a third node, wherein the main circuit is operative in response to a control signal received by the first node, and the second node is configured to receive a supply voltage; an auxiliary circuit electrically coupled to the second node of the main circuit and configured to provide surge protection for the main circuit, wherein the auxiliary circuit comprises: a second transistor, wherein a breakdown voltage of the second transistor is different than a breakdown voltage of each transistor of the plurality of first transistors, a plurality of diodes connected in serial, wherein an anode of one of the plurality of diodes is electrically coupled to a gate terminal of the second transistor, and a resistive element connected in parallel with the plurality of diodes.
2. The switching circuit of claim 1, wherein the breakdown voltage of the second transistor is greater than the breakdown voltage of each transistor of the plurality of first transistors.
3. The switching circuit of claim 1, wherein each transistor of the plurality of first transistors comprises a gate terminal electrically coupled to the first node, a drain terminal electrically coupled to the second node, and a source terminal electrically coupled to the third node.
4. The switching circuit of claim 1, wherein the auxiliary circuit further comprises: a capacitive element comprising a first terminal and a second terminal, wherein the first terminal of the capacitive element is electrically coupled to the second node, and the second terminal of the capacitive element is electrically coupled to the resistive element and the anode.
5. The switching circuit of claim 4, wherein the gate terminal of the second transistor is electrically coupled to the second terminal of the capacitive element, a drain terminal of the second transistor is electrically coupled to the second node, and a source terminal of the second transistor is electrically coupled to a ground voltage.
6. The switching circuit of claim 4, wherein a cathode of another one of the plurality of diodes is electrically coupled to a ground voltage.
7. The switching circuit of claim 1, wherein each transistor of the plurality of first transistors comprises a first high-electron-mobility transistor (HEMT) and the second transistor comprises a second HEMT.
8. The switching circuit of claim 7, wherein a substrate of the first HEMT and a substrate of the second HEMT comprise silicon.
9. The switching circuit of claim 7, wherein the first HEMT comprises a first gate structure and a first drain feature, and the first gate structure is spaced apart from the first drain feature by a first distance along a first direction, and the second HEMT comprises a second gate structure and a second drain feature, and the second gate structure is spaced apart from the second drain feature by a second distance along the first direction, wherein the second distance is greater than the first distance.
10. The switching circuit of claim 9, wherein a width of the first gate structure along a second direction is equal to a width of the second gate structure along the second direction, the second direction being substantially perpendicular to the first direction.
11. A circuit comprising: a main circuit comprising a plurality of first transistors in parallel connection and a first terminal coupled to a power line; a second transistor comprising a drain terminal electrically coupled to the first terminal of the main circuit and having a breakdown voltage higher than a breakdown voltage of the main circuit, wherein a footprint of the second

transistor is greater than a footprint of each transistor of the plurality of first transistors and is smaller than a total footprint of the plurality of first transistors; and a capacitor comprising a first terminal electrically coupled to the first terminal of the main circuit and a second terminal electrically coupled to a gate terminal of the second transistor.

12. The circuit of claim 11, wherein the breakdown voltage of the second transistor is greater than a breakdown voltage of each transistor of the plurality of first transistors.

13. The circuit of claim 12, wherein a sum of the breakdown voltage of the second transistor and the breakdown voltage of each transistor of the plurality of first transistors is greater than a surge voltage associated with the power line.

14. The circuit of claim 11, wherein an operation voltage of the second transistor is greater than an operation voltage of each transistor of the plurality of first transistors.

15. The circuit of claim 11, further comprising: a resistor comprising a third terminal electrically coupled to the second terminal and a fourth terminal electrically coupled to a ground voltage.

16. A circuit, comprising: a plurality of first transistors connected in parallel, wherein each transistor of the plurality of first transistors has a first gate structure and a first drain feature spaced from the first gate structure by a first distance; and a second transistor having a second gate structure and a second drain feature spaced from the second gate structure by a second distance the second distance is greater than the first distance, wherein the second drain feature is electrically coupled to the first drain features.

17. The circuit of claim 16, further comprising: a capacitive element comprising a first terminal and a second terminal, the first terminal of the capacitive element being electrically coupled to the first drain feature; and a resistive element comprising a third terminal and a fourth terminal, the third terminal being electrically coupled to the second terminal, and the fourth terminal being electrically coupled to a ground voltage, wherein the second gate structure is electrically coupled to the second terminal of the capacitive element, and a source feature of the second transistor is electrically coupled to the ground voltage.

18. The circuit of claim 17, further comprising: a plurality of diodes connected in serial and electrically connected between the second terminal of the capacitive element and the ground voltage.

19. The circuit of claim 16, wherein a gate width of each transistor of the plurality of first transistors is equal to a gate width of the second transistor.

20. The circuit of claim 11, wherein the second transistor comprises a III-V based high-electron-mobility transistor (HEMT).
